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I Have Covered All the Bases Here: Interpreting Reasoning Features in Large Language Models via Sparse Autoencoders

Andrey Galichin[1,2,3], Alexey Dontsov[1,5],
Polina Druzhinina[1,3], Anton Razzhigaev[1,3],
Oleg Y. Rogov[1,2,3], Elena Tutubalina[1,4], Ivan Oseledets[1,3]
[1]AIRI [2]MTUCI [3]Skoltech [4]Sber [5]HSE
Correspondence: galichin@airi.net; rogov@airi.net

Abstract

Recent LLMs like DEEPSEEK-R1 have demonstrated state-of-the-art performance by integrating deep thinking and complex reasoning during generation. However, the internal mechanisms behind these reasoning processes remain unexplored. We observe *reasoning* LLMs consistently use vocabulary associated with human reasoning processes. We hypothesize these words correspond to specific reasoning moments within the models' internal mechanisms. To test this hypothesis, we employ Sparse Autoencoders (SAEs), a technique for sparse decomposition of neural network activations into human-interpretable features. We introduce *ReasonScore*, an automatic metric to identify active SAE features during these reasoning moments. We perform manual and automatic interpretation of the features detected by our metric, and find those with activation patterns matching uncertainty, exploratory thinking, and reflection. Through steering experiments, we demonstrate that amplifying these features increases performance on reasoning-intensive benchmarks (+2.2%) while producing longer reasoning traces (+20.5%). Using the model diffing technique, we provide evidence that these features are present only in models with reasoning capabilities. Our work provides the first step towards a mechanistic understanding of reasoning in LLMs.[1]

Problem: Convert the point $(0, 3)$ in rectangular coordinates to polar coordinates. Enter your answer in the form (r, θ), where $r > 0$ and $0 \leq \theta < 2\pi$.

Generation (no steering): Okay, so I have this problem where I need to convert the rectangular coordinates (0, 3) to polar coordinates.
{... **1500** tokens ...} ← **Reasoning trace**
So, putting it all together, (0, 3) in rectangular coordinates is $\boxed{(3, \pi/2)}$ in polar coordinates.

Generation (steering): Okay, so I have this problem here where I need to convert the point (0, 3) from rectangular (which is the same as Cartesian) coordinates to polar coordinates.
{... **2000** tokens ...} ← **Increased reasoning trace**
I think I've covered all the bases here.
Calculated r, determined θ, checked using different methods, and even considered the quadrant placement. I don't see any issues with the reasoning. So, I feel confident that the polar coordinates for the point (0, 3) are $\boxed{(3, \pi/2)}$.

Figure 1: Illustration of steering (amplifying) reasoning-specific features during LLM generation. Default generation (blue) shows standard model reasoning, whereas steering (green) induces increased reasoning, self-correction, and graceful transition to the final answer—evidence that the identified features are responsible for the *reasoning* concept.

1 Introduction

Large Language Models (LLMs) have achieved remarkable success in natural language processing (Brown et al., 2020), evolving beyond simple token prediction tasks towards explicit reasoning behaviors, such as step-by-step problem-solving (Wei et al., 2022; Kojima et al., 2022; Wang et al., 2022) and self-reflection (Madaan et al., 2023; Shinn et al., 2023). Recently, specialized models which we denote as *reasoning* models, such as OpenAI's o1 (OpenAI, 2024b) and DEEPSEEK-R1 (Guo et al., 2025), have significantly improved performance on complex reasoning tasks. Trained through advanced fine-tuning and reinforcement learning (Shao et al., 2024), these models incorporate reasoning and reflective problem-solving by generating long chains of thought before providing final answers. These advances raise a new research question: How are such reasoning capabilities internally encoded within LLMs?

A growing body of work suggests that LLMs represent meaningful concepts as linear directions in their activation spaces (Mikolov et al., 2013; Elhage et al., 2022; Park et al., 2023; Nanda et al.,

[1]Code available at https://github.com/AIRI-Institute/SAE-Reasoning

2023; Jiang et al., 2024). However, identifying these directions remains challenging. Sparse Autoencoders (SAEs) offer a principled approach to disentangle activations into sparse, interpretable *features* (Cunningham et al., 2023; Gao et al., 2024b; Templeton, 2024; Marks et al., 2024). Given a trained SAE, the interpretation of its features could be performed by activation analysis (Bricken et al., 2023), targeted interventions (Templeton, 2024), or automated methods (Paulo et al., 2024; Kuznetsov et al., 2025). While SAEs have proven effective in discovering features for various concepts (Shu et al., 2025), their ability to isolate reasoning-specific features remains unexplored.

In this work, we investigate whether reasoning processes in *reasoning* LLMs can be identified and decomposed into interpretable directions within their activation spaces. We analyze the outputs produced by these models', and find a consistent pattern in which they employ words associated with human reasoning processes: uncertainty (e.g. "perhaps"), reflection (e.g. "however"), and exploration (e.g. "alternatively") (Chinn and Anderson, 1998; Boyd and Kong, 2017; Gerns and Mortimore, 2025). We hypothesize that these linguistic patterns correspond to the moments of reasoning within the models' internal mechanisms. To test this, we construct a vocabulary of reasoning words. We then use SAEs to decompose LLM activations into interpretable features and propose `ReasonScore`, a metric that quantifies the degree to which a given SAE feature is active on the reasoning vocabulary.

We evaluate the features found by `ReasonScore` using manual (Bricken et al., 2023) and automatic interpretation (Kuznetsov et al., 2025) techniques, and find the set of 46 features that demonstrate interpretable activation patterns corresponding to uncertainty, exploratory thinking, and reflection. We perform steering experiments and show that amplifying these reasoning features leads to improved performance on reasoning-intensive benchmarks ($+13.4\%$ on AIME-2024, $+2.2\%$ on MATH-500, and $+4\%$ on GPQA Diamond) while producing longer reasoning traces ($+18.5\%$ on AIME-2024, $+20.5\%$ on MATH-500, and $+13.9\%$ on GPQA Diamond). Through model diffing (Bricken et al., 2024), we demonstrate that these reasoning features emerge only in *reasoning* LLMs and are absent in base models. Our results provide mechanistic evidence that specific, interpretable components in LLMs representations are causally linked to reasoning behavior.

The contributions of this paper are the following:

- We introduce `ReasonScore`, an automatic metric to identify the SAE features responsible for reasoning and confirm its effectiveness using interpretability techniques.
- We provide causal evidence from steering experiments, demonstrating that amplifying identified features induces reasoning behavior.
- We analyze the emergence of reasoning features in LLMs through model diffing technique, and confirm their existence only after the *reasoning* fine-tuning stage.

2 Interpretability with SAEs

Sparse Autoencoders (SAEs) aim to learn a sparse decomposition of model activations to identify disentangled features that correspond to meaningful concepts (Bricken et al., 2023). Here, a *feature* refers to an individual component of the learned representation that captures specific, human-interpretable characteristics of the input data.

The core idea behind SAEs is to reconstruct model activations $x \in \mathbb{R}^n$ as a sparse linear combination of learned feature directions, where the feature *dictionary* dimensionality $m \gg n$. Formally, we extract LLM activations from some intermediate state in the model and train a two-layer autoencoder:

$$\begin{aligned} f(x) &= \sigma(W_{\text{enc}} x + b_{\text{enc}}), \\ \hat{x}(f) &= W_{\text{dec}} f + b_{\text{dec}}. \end{aligned} \tag{1}$$

Here, $f(x) \in \mathbb{R}^m$ is a sparse vector of feature magnitudes and $\hat{x}(f) \in \mathbb{R}^n$ is a reconstruction of the original activation x. The columns of W_{dec}, which we denote by $W_{\text{dec},i}$, $i = 1, \ldots, m$, represent the dictionary of directions, or *features*, into which the SAE decomposes x. The activation function σ enforces non-negativity in $f(x)$.

The training objective used to train Sparse Autoencoders minimizes a reconstruction loss $\mathcal{L}_{\text{recon}}$ and an additional sparsity-promoting loss $\mathcal{L}_{\text{sparsity}}$. This objective forces SAE to learn a small set of interpretable features that capture the distinct properties of the activations.

In our work, we use vanilla SAE (Bricken et al., 2023) with `ReLU` activation function. Following (Conerly et al., 2024), we use a squared error reconstruction loss and a modified L1 penalty as a

sparsity loss:

$$\mathcal{L} = \underbrace{\|x - \hat{x}\|_2^2}_{\mathcal{L}_{\text{recon}}} + \lambda \underbrace{\sum_{i=1}^{m} f_i \|W_{\text{dec},i}\|_2}_{\mathcal{L}_{\text{sparsity}}}, \quad (2)$$

where λ is the sparsity penalty coefficient.

3 Method

We identify reasoning-specific features through a two-step approach. First, we examine the language space of reasoning words used by *reasoning* LLMs, and construct the respective vocabulary $\mathcal{R}$ (Sec. 3.1). Secondly, we introduce `ReasonScore` to find the sparse autoencoder features responsible for reasoning capabilities (Sec. 3.2).

3.1 Reasoning Vocabulary

Reasoning words are linguistic features associated with exploratory talk as humans talk-to-learn, explore ideas, and probe each other's thinking (Boyd and Kong, 2017).

In the original DEEPSEEK-R1 paper (Guo et al., 2025), the authors demonstrated that the model spontaneously exhibits sophisticated human-like behaviors, such as reflection, where it revisits and reevaluates its previous steps, and exploration of alternative problem-solving approaches. In particular, the model explicitly employs words that mirror the introspective language humans use when thinking (such as "maybe", "but", "wait"). We hypothesize that these moments correspond directly to the internal reasoning process of the models, which is consistent with studies on human thinking (Chinn and Anderson, 1998; Boyd and Kong, 2017).

To extract the models' reasoning vocabulary, we use an approach similar to that of (Rayson and Garside, 2000). We construct two corpora from the OPENTHOUGHTS-114K (OpenThoughts, 2025) dataset: ground-truth samples containing formal and step-by-step solutions to the problems, and the solutions obtained using DEEPSEEK-R1 for the same problems. For each word, we calculate its frequency in the tasks solutions p_{solution} and in the thinking solutions p_{think}, then sort all words by the frequency difference $p_{\text{think}} - p_{\text{solution}}$. Next, we select the top-k words by frequency difference, where k is determined by the point where the frequency distribution plateaus, and filter out words with high presence in the Google Books Ngram Corpus (Michel et al., 2011) (Fig. 2). To determine the final vocabulary from this candidate set, we choose words that best capture reasoning behavior.

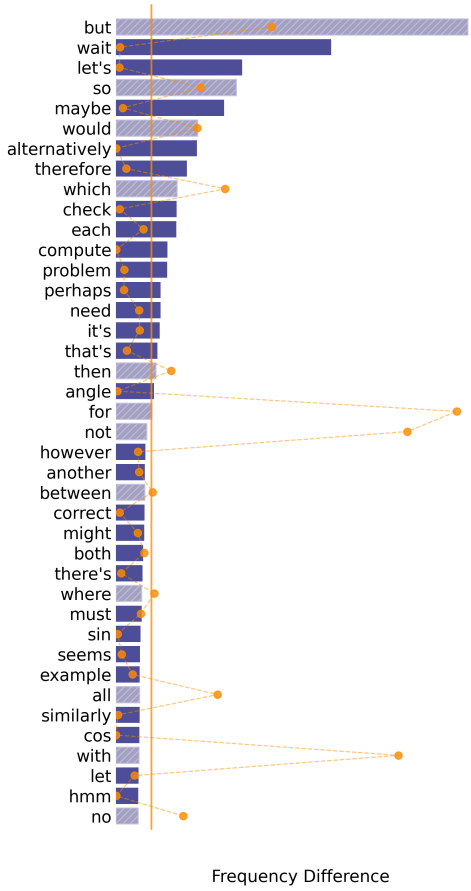

Figure 2: The distribution of top 40 words with the greatest change in frequency between reasoning traces of DEEPSEEK-R1 and ground-truth solutions of math problems. Orange dots show the frequency from Google Books Ngram Corpus. We remove the words with absolute frequency above the pre-defined threshold (orange line), and keep those with the high relative frequency indicating reasoning.

This includes words that match those considered in the linguistic literature (Chinn and Anderson, 1998; Boyd and Kong, 2017) and those that we identify through manual analysis of model traces exhibiting reasoning patterns.

Following this pipeline, we select 10 words indicating reasoning as models' *reasoning* vocabulary and denote it by $\mathcal{R}$. The exact list of words can be found in Appx. A.1. Ablation experiments confirm that these words play a functional role in reasoning capabilities (see Sec. 4.3 for setup, Appx. A.2 for results).

3.2 ReasonScore

To find SAE features that capture reasoning-related behavior, we follow our hypothesis and introduce `ReasonScore`, which measures the contribution